From Quantum Shots to Training Data: Reorganizing Measurement Records in Quantum Machine Learning

Markus Baumann*, Maximilian Zorn*, Thomas Gabor†, Claudia Linnhoff-Popien*, and Jonas Stein*
*QAR-Lab, Department of Computer Science, LMU Munich, Munich, Germany
†Department of Computer Science, University of Exeter, Exeter, United Kingdom

***Abstract*—Between keeping every shot as a noisy training example and averaging all shots into one clean feature lies an entire spectrum of data organizations that quantum machine learning usually leaves implicit. Shot grouping makes this choice explicit by partitioning a fixed measurement record into disjoint groups and averaging within each group, tuning smoothly between the two conventions with a single validated parameter and no additional quantum executions. We evaluate the method on chaotic synthetic benchmarks and on real-world data under strict execution budgets. At the balanced allocation, intermediate grouping lowers mean error across the evaluated tasks, with supported improvements on every task. Matched controls show that the gain acts as correctly scaled regularization of the classical readout. Unmitigated runs on two superconducting processors show positive mean differences on every task. Shot grouping therefore offers a practical way to improve finite-shot quantum learning at no added quantum cost.**

***Index Terms*—Quantum computing, Machine learning, Reservoir computing, Time series analysis**

I. Introduction

Every quantum learning experiment ends the same way: a circuit runs many times, and each run returns one measured bitstring. Before anything can be learned, this pile of shots must become training data. The step looks like bookkeeping, and it is usually treated that way. Almost every pipeline averages all shots of a time step into a single expectation-value estimate and moves on. This paper argues that the step is not bookkeeping. It is a statistical decision with measurable consequences, and treating it as a validated hyperparameter improves finite-shot quantum learning at no additional quantum cost.

The averaging convention is well motivated, because expectation values are the quantities predicted by theory and estimated in experiments. A learning pipeline, however, does not consume observables. It consumes training examples, and it cares about two things only: how many there are and how noisy each one is. The moment a measurement record becomes training data, a physics convention quietly turns into a statistical choice: spend all measurements on making one example clean, or make several examples that are merely clean enough. No single choice can suit every task, budget, and noise level.

Averaging everything is one end of a dial. It buys the cleanest possible feature vector but pays with scarcity: thousands of circuit executions collapse into a handful of training examples, and too few examples relative to the number of features destabilizes regression [1]. The opposite end, keeping every shot as its own example, buys abundance but pays with noise, and noisy features systematically shrink fitted coefficients [2]. Between these extremes lies an entire range of aggregations that quantum machine learning usually leaves implicit.

Corresponding author: markus.baumann@campus.lmu.de.

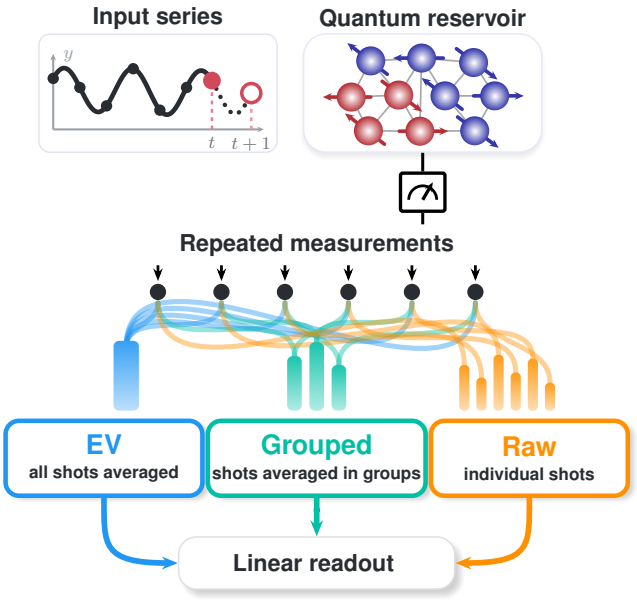


Fig. 1. Three ways to turn one shot record into training data. A fixed quantum reservoir is measured many times per step, and only the classical step changes: EV averages all shots, Raw keeps each shot, and Grouped averages disjoint groups. All feed the same linear readout.

Adaptive shot aggregation makes this choice explicit. The N shots of each time step are partitioned into disjoint groups of size k and averaged within each group, so that a single parameter slides smoothly from raw shots ($k = 1$) to full expectation-value averaging ($k = N$). The group size is a validation-selected hyperparameter of the classical readout, chosen by time-respecting validation jointly with the readout regularization. The procedure operates purely on the classical record: it requires no new circuits, no extra shots, and it can be applied retroactively to any archived record in which individual outcomes were retained.

Quantum reservoir computing (QRC) is a clean place to study this dial [3]–[5]. A fixed quantum circuit transforms an input

sequence into measured features, and only a classical readout is trained. The circuit, the targets, and the measurement record can therefore be held completely fixed while the aggregation alone changes. Whatever happens to the forecast is attributable to the aggregation, because nothing else moved. We compare three representations of one record (Fig. 1): the expectation-value representation (EV), individual shots (Raw), and the grouped representation in between.

We find that the aggregation scale materially affects test error under fixed execution budgets: at the balanced allocation, an intermediate scale lowers mean test error relative to full averaging on all four evaluated forecasting tasks, with paired confidence intervals above zero in every case. The best scale depends on the record, so no single convention is appropriate. Matched controls trace the gain to covariance regularization of the classical readout, supplied automatically by the record's own shot fluctuations, and the effect transfers to unmitigated experiments on real hardware. Our claims concern how to organize a given measurement record, not quantum advantage.

In summary, this paper makes the following contributions:

- **A tunable aggregation procedure.** We make k a validation-selected classical-readout hyperparameter spanning Raw to EV, chosen jointly with ridge strength and applicable retroactively at no added quantum cost (Sec. III).
- **Evidence that the choice matters.** Under a fixed budget of 10,000 executions, intermediate aggregation lowers mean error on all four tasks; a three-allocation sweep shows that gains are largest with many shots at few time steps (Secs. V-A and V-B).
- **A mechanism.** An exact linear-readout identity equates grouped training with EV training plus a shot-covariance penalty; the matched Regularized EV control confirms this mechanism (Secs. III-C and V-C).
- **Transfer to hardware.** Unmitigated experiments with a separate ten-qubit reservoir on two processors show positive mean improvements on every task (Sec. V-D).

II. Related Work

Finite sampling can be addressed at three stages of a quantum learning pipeline: before execution, by allocating the measurement budget; after averaging, by correcting expectation-value estimates; or at the measurement record itself, by deciding how repeated outcomes become training examples. Our method focuses on this third stage.

A brief instance of subset averaging appears in Cemin *et al.* [6], where it is used as an implementation heuristic to enlarge the training set for reduced quantum channel learning. We develop this idea into a tunable procedure: the group size spans the full range from individual shots to complete expectation-value averaging, is selected jointly with readout regularization, and is evaluated under fixed execution budgets. For linear readouts, grouping also has an exact regularization interpretation, connecting it to classical work on training with noisy inputs [7].

Complementary work addresses the other stages of finite-shot learning. Resource-aware optimization allocates measurements before execution [8]. After averaging, noise-aware training rules [9] and variance regularization [10] reduce the effect of sampling noise, while fundamental limits for learning from finitely sampled physical systems have also been studied [11]. Shot grouping instead reorganizes an existing measurement record. It requires no additional quantum executions and can be applied retrospectively whenever individual outcomes have been retained.

Within QRC, prior work has examined how noise, dissipation, measurement, finite Hilbert-space dimension, and input transformations shape temporal features [12]–[23]. We hold the reservoir and measurement record fixed and change only how repeated outcomes are presented to the classical readout. Shot grouping is therefore complementary to these approaches and is not tied to a particular reservoir design.

III. Method

A. Adaptive shot aggregation

Let $x_{t,n}$ denote the features produced by shot n at time t. For each time step, draw uniformly at random a partition $\{\mathcal{I}_g\}_{g=1}^{G}$ of the N shot indices into $G = N/k$ disjoint blocks of equal size k, independently across time steps, and define

$$\bar{x}_t^{(g)} = \frac{1}{k} \sum_{n \in \mathcal{I}_g} x_{t,n}, \qquad g = 1, \ldots, G. \tag{1}$$

Every $\bar{x}_t^{(g)}$ is a partly averaged view of the same time step, and all views share the same target y_t. At test time we average the G predictions of a step, which is identical to predicting once from the averaged features. The two methods are therefore scored in exactly the same way, so any difference between EV and Grouped is created during training, not at test time.

The shot index carries no physical meaning, because shot n at one step and shot n at the next come from unrelated circuit runs. We repeat the whole analysis over several independent partition draws.

B. Fixed execution budget and training

Evaluating T time steps with N shots per step costs TN circuit executions. We hold this budget fixed at ten thousand executions and study three ways of spending it, $(T, N) = (250, 40)$, $(500, 20)$, and $(1000, 10)$, ranging from fewer time steps with more shots to more time steps with fewer shots. For each setting we try every group size k that divides N, which includes Raw at $k = 1$ and EV at $k = N$.

Selection and testing respect time order. After a short washout, the earlier part of each series is the selection data used to choose settings, and the final fifth is held out as the test block and left untouched until the end. On the selection data we pick the group size and ridge strength (k, λ) by forward-chaining validation. In three successive rounds we fit on an earlier stretch and score on the stretch that immediately follows, so training always precedes validation in time, and each candidate pair is ranked by its validation error averaged over the three rounds.

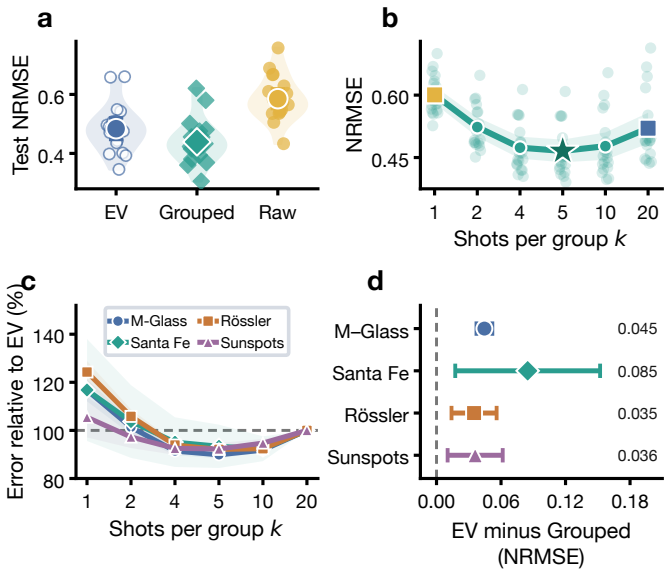


Fig. 2. Adaptive aggregation under the shared protocol. (a) Test-error distributions for EV, Grouped, and Raw across twenty Mackey–Glass trajectories. (b) Error is U-shaped in k; the star marks the validated choice. (c) The task means show the same dip below EV. (d) Paired EV-minus-Grouped test improvements; positive values favor Grouped. Large markers and bars are means and 95% intervals. Panels (b,c) use development data and (a,d) test data.

The best pair is then frozen, the readout is refitted on the whole selection data, and the test block is scored once. Normalization and feature scaling are computed from training data only, so the test block never influences any choice.

C. Why grouping regularizes the readout

Full averaging gives the readout one relatively stable feature vector per time step. Raw gives it many noisier vectors. Grouping lies between them, providing several partly averaged views of the same time step with the same target.

Stable feature combinations should predict similarly across these views, whereas shot-sensitive combinations do not. Fitting every view to the same target therefore discourages reliance on finite-shot fluctuations.

For a linear readout with squared loss, this effect is exact. Let $\bar{x}_t = G^{-1}\sum_{g=1}^{G} \bar{x}_t^{(g)} = N^{-1}\sum_{n=1}^{N} x_{t,n}$ denote the fully averaged (EV) features, and let

$$C_{k,t} = \frac{1}{G}\sum_{g=1}^{G}\left(\bar{x}_t^{(g)} - \bar{x}_t\right)\left(\bar{x}_t^{(g)} - \bar{x}_t\right)^{\top}$$

be the covariance of the grouped views around that average. For any intercept b and weights w,

$$\frac{1}{G}\sum_{g=1}^{G}\left(y_t - b - w^{\top}\bar{x}_t^{(g)}\right)^2 = \left(y_t - b - w^{\top}\bar{x}_t\right)^2 + w^{\top}C_{k,t}w. \quad (2)$$

The first term is the EV loss. The second penalizes feature combinations that fluctuate across views. Grouped training is therefore EV training with a regularizer obtained from observed shot fluctuations. The identity requires neither Gaussian noise nor independent time steps.

On a fixed common feature scale, let $S_{\mathrm{shot},t}$ be the $1/N$ covariance of the N shot-level features. Before representation-specific standardization, averaging over random partitions $\mathcal{P}$ gives

$$\mathbb{E}_{\mathcal{P}}[C_{k,t}] = \frac{N-k}{k(N-1)}S_{\mathrm{shot},t}. \quad (3)$$

The multiplier is one at $k = 1$ and zero at $k = N$. Thus, k mainly controls regularization strength: smaller groups retain more shot variation, while larger groups average it away. A finite partition can also change the realized penalty, especially when only a few groups are available.

This motivates *covariance-regularized expectation-value training (Regularized EV)*, which retains the fully averaged EV features but adds a penalty estimated from shot-level fluctuations, on top of the ridge term that all methods share. Its covariance and ridge strengths are selected on the same validation data, independently of Grouped's k. Similar performance from Grouped and Regularized EV would support covariance regularization as the mechanism behind the gain.

As a check on our implementation, at a fixed group size and ridge strength we compute the Grouped fit in two mathematically equal ways. The first fits ridge directly on all grouped training views. The second uses the two-term form of Eq. (2), fitting on the mean view with the added covariance penalty. With the same feature standardization, the two agree to numerical precision, which confirms that the decomposition is implemented correctly.

IV. Experimental Setup

We run two studies. The first is a controlled simulation on four forecasting tasks. The second repeats the core comparison on real quantum hardware. In both, every aggregation choice reads the very same measurement record, so any difference in accuracy comes only from how the shots are organized and not from running the circuit differently.

A. Simulation reservoir and evaluation protocol

The simulated reservoir is a fixed six-qubit circuit with two shallow data reuploading layers [24]. Each layer applies input-dependent R_y and R_z rotations with fixed per-qubit offsets, nearest-neighbor R_{zz} couplings, and fixed non-Clifford mixing rotations. For each circuit setting, all angles are sampled once and then held fixed; three frozen settings are used to assess sensitivity to this draw. Each computational-basis shot yields 11 features: six local observables Z_i and five nearest-neighbor products Z_iZ_{i+1}. A leaky integrator with leak rate 0.2 and a ten-step lag window provide temporal memory, and a ridge-regression readout predicts the next value.

We evaluate one-step-ahead forecasting on four time series: the synthetic Mackey–Glass [25] and Rössler [26] systems, and the measured Santa Fe laser intensity [27] and monthly sunspot number [28]. All tasks use the same reservoir architecture, feature construction, temporal processing, scaling, validation procedure, and execution budgets. The same selection rule is applied separately to each task; no task-specific changes are made to the reservoir or preprocessing pipeline.

All four finite-shot methods use the same measurement record. Two additional controls use different feature sources: an ideal reservoir without finite-shot sampling noise and a classical baseline that applies the same temporal processing and readout directly to the input series. These controls distinguish the effect of aggregation from the effects of finite sampling and of the reservoir transformation itself.

The trajectory or time window is the unit of uncertainty estimation. Mackey–Glass uses twenty independently generated trajectories, each with a held-out test block. Rössler uses five trajectories, and each real-world task uses five windows with disjoint test blocks. Within each unit, we repeat the analysis over three circuit settings, three measurement draws, and three random groupings where applicable. These repeats are averaged within the unit before paired confidence intervals are formed across the independent units for that task. Grouped views, neighboring time steps, and repeated evaluations of the same sequence are not treated as independent observations, because they share measurements or are coupled by the leaky integrator and overlapping lags.

Model selection uses no test data. After a short washout, the final 20% of each trajectory or window is reserved as an untouched test block. The preceding data are used in three expanding-window validation splits, each trained on the available past and evaluated on the immediately following block. The group size k and ridge strength λ are selected from these validation results. The readout is then refitted on all data preceding the test block and evaluated once on the reserved block. For representations with multiple views at one time step, the corresponding predictions are averaged to produce one forecast, so all methods are scored on the same targets.

The primary budget uses $T = 500$ time steps and $N = 20$ shots per step, giving 10,000 circuit executions. We also test $(T, N) = (250, 40)$ and $(1000, 10)$, which spend the same budget with different balances between observed time steps and shots per step. All reservoir parameters, data-generation settings, window origins, and selection rules are fixed and stored before any test block is evaluated.

B. Hardware setup

The hardware study uses a different reservoir: a fixed ten-qubit circuit with one input qubit, nine memory qubits, a ring of entangling gates, and ten local outputs mapped on *ibm_fez* and *ibm_marrakesh*. It tests transfer of the aggregation procedure; it is not a hardware implementation of the six-qubit simulator or a comparison of qubit counts. Every method receives the same circuit outputs. For each task and processor, one acquisition of 400 shots at 500 time steps is split into twenty ordered, disjoint records of 20 shots. The intervals measure variation within that acquisition, not across devices or calibration periods.

V. Evaluation

We report accuracy as test NRMSE, a normalized error for which lower is better and where one means predicting the mean. We summarize the benefit of Grouped as the paired difference EV minus Grouped, so a positive value means Grouped is more accurate. Each difference comes with a 95% confidence interval: an interval above zero provides evidence of a positive mean improvement, while an interval containing zero is inconclusive. The balanced setting spends the fixed budget on 500 time steps with 20 shots each.

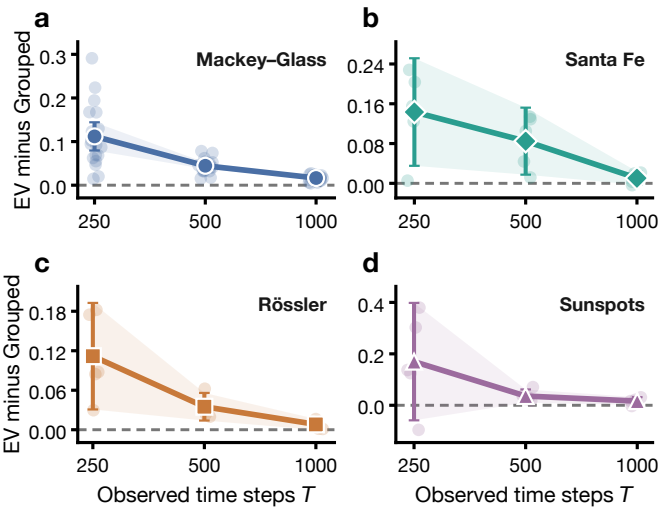


Fig. 3. Fixed-budget allocation. Panels show EV-minus-Grouped NRMSE, so positive values favor Grouped. Moving right spends the budget on more steps and fewer shots; gains are largest with many shots and few steps. Markers and bars are means and 95% intervals.

TABLE I
MEAN TEST NRMSE AT THE BALANCED SETTING. REPEATED MEASUREMENT TRIALS ARE AVERAGED WITHIN INDEPENDENT TRAJECTORIES OR WINDOWS. LOWER VALUES ARE BETTER.

Task	Class.	Ideal	EV	Grouped	Raw	Regularized EV
Mackey–Glass	0.098	0.190	0.484	0.439	0.586	**0.430**
Santa Fe laser	0.762	0.493	0.764	0.680	0.697	**0.677**
Rössler	0.096	0.054	0.414	0.379	0.580	**0.367**
Sunspots	0.387	0.474	0.553	0.517	0.558	**0.515**

A. Balanced setting

At the balanced budget, Grouped lowers mean NRMSE relative to EV on all four tasks (Table I, Fig. 2). For Mackey–Glass, the primary estimate uses twenty independent held-out trajectories. The improvement is 0.0445 on Mackey–Glass (95% CI $[0.0372, 0.0519]$), 0.0848 on Santa Fe laser ($[0.0174, 0.1522]$), 0.0349 on Rössler ($[0.0139, 0.0559]$), and 0.0360 on sunspots ($[0.0103, 0.0616]$). Every interval stays above zero.

Santa Fe laser shows the largest effect. Grouped reduces the error from 0.764 to 0.680, and all five disjoint test windows favor Grouped. When k and the ridge strength are tuned together, the search lands on $k = 10$ most often and on an intermediate scale, neither Raw nor EV, in 94% of repeated trials. Regularized EV reaches 0.677, almost the same as Grouped, which already suggests that the gain is a regularization effect rather than new physical information. The gain also depends on how the budget is spent. On Santa Fe it shrinks from 0.143 with 40 shots per step to 0.010 with 10

shots per step because grouping has more shots to reorganize when shots are plentiful.

We also compare an ideal noise-free reservoir with a classical baseline that uses identical preprocessing. The quantum reservoir is worse on Mackey–Glass and sunspots, better on Santa Fe, and inconclusive on Rössler. Our claims are therefore about organizing a chosen reservoir record, not quantum advantage. The preferred aggregation scale varies across tasks, trajectories, circuit settings, measurement draws, and grouping assignments, and EV is occasionally best. On a fixed feature scale, Eq. (3) clarifies the role of k: it mainly sets the strength of the covariance penalty, while a finite random grouping can also change the realized penalty. Raw is less accurate than Grouped on all four task means.

B. Trading time steps against shots

Figure 3 compares three ways of spending the fixed budget. With 40 shots at each of 250 time steps, Grouped improves every task by 0.112 to 0.170. The interval stays above zero for Mackey–Glass, Santa Fe laser, and Rössler, while the sunspot result varies across windows. When the same budget is spread over 1000 time steps with only 10 shots each, the gains shrink to between 0.008 and 0.016, and the interval stays above zero only for Mackey–Glass and sunspots.

C. Why the gain is regularization

Regularized EV keeps EV features and adds a penalty based on how the shots fluctuate together. At the balanced setting its mean error is 0.0021 to 0.0116 NRMSE below Grouped. The Mackey–Glass and Rössler intervals exclude zero. This near-match supports the regularization mechanism in Eq. (2) rather than extra device information (Fig. 4). Raw is less accurate than EV on Mackey–Glass and Rössler and inconclusive on the two real-world datasets. At fixed k, the exact two-term loss reproduces the Grouped ridge fit to numerical precision. On a fixed feature scale, Eq. (3) shows how the induced penalty changes with k, whereas Regularized EV tunes the full shot-covariance penalty directly.

D. Transfer to hardware

Figure 5 reports the unmitigated hardware results. Grouped improves every task, with mean gains from 0.038 to 0.153 NRMSE on *ibm_fez* and from 0.031 to 0.103 on *ibm_marrakesh*. On Santa Fe laser, the error falls from 0.750 to 0.711 on *ibm_fez* and from 0.744 to 0.713 on *ibm_marrakesh*. Grouped is more accurate in 19 of 20 measurement blocks on each processor.

VI. Discussion

A quantum experiment produces a measurement record, not a ready made training set, so organizing repeated shots is part of the learning algorithm. Adaptive shot aggregation changes only this classical organization: the circuit and record remain fixed, settings are chosen before testing, and gains arise during training at no additional quantum cost.

The choice matters most when many shots are collected at few time steps: full averaging then yields few precise examples,

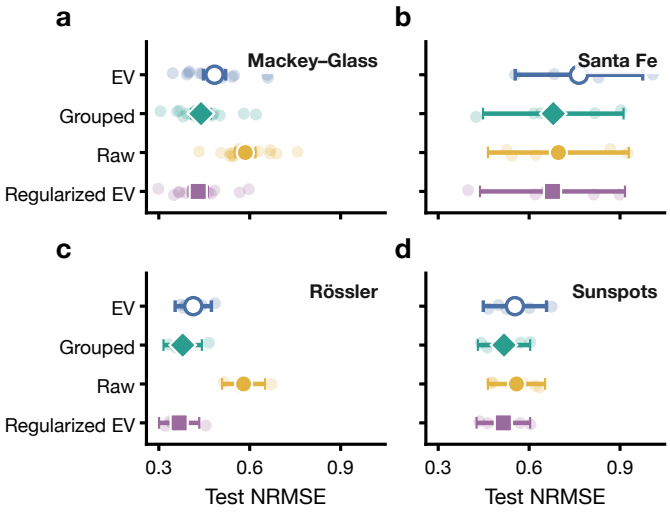


Fig. 4. Balanced-setting controls, lower is better. Grouped and Regularized EV are nearly equal and lowest on every task, supporting a readout-regularization explanation; EV and Raw trail. Faint points are independent-unit means, and markers and bars are means and 95% intervals.

whereas grouping creates several moderately denoised views whose differences reveal stable feature directions. With fewer shots per step, both the freedom to reorganize the record and its benefit shrink. The aggregation scale should therefore be validated rather than fixed by convention.

For a linear readout, this effect has an exact interpretation. Equation (2) separates the grouped loss into the EV loss and a quadratic covariance penalty. Since every view shares the

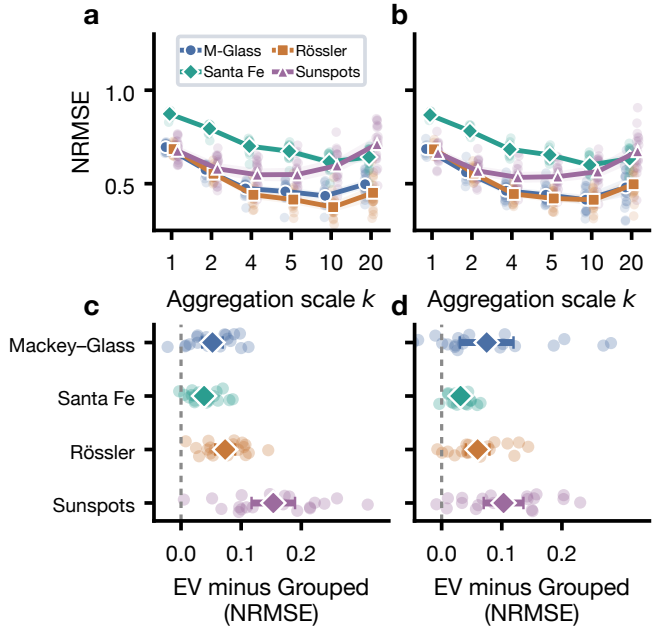


Fig. 5. Transfer to two unmitigated processors. (a,b) On *ibm_fez* and *ibm_marrakesh* the error is again U-shaped in the aggregation scale k. (c,d) The improvement of Grouped over EV is positive on all four tasks and both devices. One $T = 500$ acquisition of 400 shots per step is split into twenty disjoint $N = 20$ records, and intervals describe variation within that single acquisition.

same target, the penalty discourages large weights on feature combinations that vary between groups. On a common feature scale, Eq. (3) shows that smaller groups produce stronger regularization and $k = N$ recovers EV.

Regularized EV's near-match to Grouped supports this explanation. It tunes the covariance penalty directly, whereas Grouped selects among strengths induced by the available group sizes. Grouping is therefore a simple route to this regularization, but not the only one.

Individual shot outcomes should be retained. Once a record has been reduced to expectation values, alternative aggregation scales cannot be reconstructed. Keeping the full record preserves this option without repeating the quantum experiment.

The effect transfers without mitigation to a separate ten-qubit circuit on two processors, where Grouped improves on EV across all tasks. The intervals cover blocks within one acquisition per task and processor, not reproducibility across runs, calibrations, or days. Independent hardware campaigns, other reservoirs, and richer readouts are the next tests.

VII. Conclusion

Turning a shot record into training data has always involved a choice, and the field has usually answered it by averaging everything. Choosing the aggregation scale by validation lowers mean test error on all four forecasting tasks at the balanced budget, with paired confidence intervals above zero in every task. The hardware study shows positive mean differences on both superconducting processors, and the method requires no additional quantum cost. Our practical recommendation is therefore simple: retain individual shot outcomes, and select the group size k jointly with the readout regularization by validation instead of fixing it by convention. Every measurement-based reservoir produces such a record, so this choice is available to all of them. The underlying question, how to turn finitely sampled noisy measurements into training data, reaches beyond reservoirs to finite-shot learning as a whole.

Appendix A
Reproducibility

The supplement and repository provide complete circuit and hardware metadata, hyperparameter grids, normalization and interval procedures, random seeds, data windows, records, and code for every figure and table. The named artifact is available at github.com/eybmits/adaptive-shot-aggregation.

References

[1] D. Hsu, S. M. Kakade, and T. Zhang, "Random Design Analysis of Ridge Regression," in *Proceedings of the 25th Conference on Learning Theory*, ser. Proceedings of Machine Learning Research, vol. 23, 2012, pp. 9.1–9.24.

[2] W. A. Fuller, *Measurement Error Models*. New York: Wiley, 1987.

[3] K. Fujii and K. Nakajima, "Harnessing Disordered-Ensemble Quantum Dynamics for Machine Learning," *Physical Review Applied*, vol. 8, no. 2, p. 024030, 2017.

[4] K. Nakajima, K. Fujii, M. Negoro, K. Mitarai, and M. Kitagawa, "Boosting Computational Power through Spatial Multiplexing in Quantum Reservoir Computing," *Physical Review Applied*, vol. 11, no. 3, p. 034021, 2019.

[5] P. Mujal, R. Martínez-Peña, J. Nokkala, J. García-Beni, G. L. Giorgi *et al.*, "Opportunities in Quantum Reservoir Computing and Extreme Learning Machines," *Advanced Quantum Technologies*, vol. 4, no. 8, p. 2100027, 2021.

[6] G. Cemin, M. Cech, E. Weiss, S. Soltan, D. Braun *et al.*, "Machine Learning of Reduced Quantum Channels on Noisy Intermediate-Scale Quantum Devices," *Physical Review A*, vol. 110, no. 5, p. 052418, 2024.

[7] C. M. Bishop, "Training with noise is equivalent to Tikhonov regularization," *Neural Computation*, vol. 7, no. 1, pp. 108–116, 1995.

[8] C. Moussa, M. H. Gordon, M. Baczyk, M. Cerezo, L. Cincio, and P. J. Coles, "Resource Frugal Optimizer for Quantum Machine Learning," *Quantum Science and Technology*, vol. 8, no. 4, p. 045019, 2023.

[9] O. Ahmed, F. Tennie, and L. Magri, "Optimal Training of Finitely Sampled Quantum Reservoir Computers for Forecasting of Chaotic Dynamics," *Quantum Machine Intelligence*, vol. 7, p. 31, 2025.

[10] D. A. Kreplin and M. Roth, "Reduction of Finite Sampling Noise in Quantum Neural Networks," *Quantum*, vol. 8, p. 1385, 2024.

[11] F. Hu, G. Angelatos, S. A. Khan, M. Vives, E. Türeci *et al.*, "Tackling Sampling Noise in Physical Systems for Machine Learning Applications: Fundamental Limits and Eigentasks," *Physical Review X*, vol. 13, no. 4, p. 041020, 2023.

[12] T. Kubota, Y. Suzuki, S. Kobayashi, Q. H. Tran, N. Yamamoto, and K. Nakajima, "Quantum Noise-Induced Reservoir Computing," 2022.

[13] A. Palacios de Luis, R. Martínez-Peña, M. C. Soriano, G. L. Giorgi, and R. Zambrini, "Role of Coherence in Many-Body Quantum Reservoir Computing," *Communications Physics*, vol. 7, no. 1, p. 369, 2024.

[14] T. Kubota, Y. Suzuki, S. Kobayashi, Q. H. Tran, N. Yamamoto, and K. Nakajima, "Temporal Information Processing Induced by Quantum Noise," *Physical Review Research*, vol. 5, no. 2, p. 023057, 2023.

[15] R. Martínez-Peña and J.-P. Ortega, "Quantum Reservoir Computing in Finite Dimensions," *Physical Review E*, vol. 107, no. 3, p. 035306, 2023.

[16] P. Mujal, R. Martínez-Peña, G. L. Giorgi, M. C. Soriano, and R. Zambrini, "Time-Series Quantum Reservoir Computing with Weak and Projective Measurements," *npj Quantum Information*, vol. 9, p. 16, 2023.

[17] Y. Suzuki, Q. Gao, K. C. Pradel, K. Yasuoka, and N. Yamamoto, "Natural Quantum Reservoir Computing for Temporal Information Processing," *Scientific Reports*, vol. 12, p. 1353, 2022.

[18] A. Sannia, R. Martínez-Peña, M. C. Soriano, G. L. Giorgi, and R. Zambrini, "Dissipation as a Resource for Quantum Reservoir Computing," *Quantum*, vol. 8, p. 1291, 2024.

[19] L. C. G. Govia, G. J. Ribeill, G. E. Rowlands, and T. A. Ohki, "Nonlinear Input Transformations Are Ubiquitous in Quantum Reservoir Computing," *Neuromorphic Computing and Engineering*, vol. 2, no. 1, p. 014008, 2022.

[20] R. Martínez-Peña, J. Nokkala, G. L. Giorgi, R. Zambrini, and M. C. Soriano, "Information Processing Capacity of Spin-Based Quantum Reservoir Computing Systems," *Cognitive Computation*, vol. 15, no. 5, pp. 1440–1451, 2023.

[21] W. D. Kalfus, G. J. Ribeill, G. E. Rowlands, H. K. Krovi, T. A. Ohki, and L. C. G. Govia, "Hilbert Space as a Computational Resource in Reservoir Computing," *Physical Review Research*, vol. 4, no. 3, p. 033007, 2022.

[22] D. Fry, A. Deshmukh, S. Y.-C. Chen, V. Rastunkov, and V. Markov, "Optimizing Quantum Noise-Induced Reservoir Computing for Nonlinear and Chaotic Time Series Prediction," *Scientific Reports*, vol. 13, p. 19326, 2023.

[23] L. C. G. Govia, G. J. Ribeill, G. E. Rowlands, H. K. Krovi, and T. A. Ohki, "Quantum Reservoir Computing with a Single Nonlinear Oscillator," *Physical Review Research*, vol. 3, no. 1, p. 013077, 2021.

[24] A. Pérez-Salinas, A. Cervera-Lierta, E. Gil-Fuster, and J. I. Latorre, "Data Re-Uploading for a Universal Quantum Classifier," *Quantum*, vol. 4, p. 226, 2020.

[25] M. C. Mackey and L. Glass, "Oscillation and Chaos in Physiological Control Systems," *Science*, vol. 197, no. 4300, pp. 287–289, 1977.

[26] O. E. Rössler, "An equation for continuous chaos," *Physics Letters A*, vol. 57, no. 5, pp. 397–398, 1976.

[27] N. A. Gershenfeld and A. S. Weigend, "The Future of Time Series," Santa Fe Institute, Tech. Rep. 93-08-053, 1993.

[28] SILSO World Data Center, "Monthly mean total sunspot number V2.0," 2026.